

TradeM8 - Your Trading Assistant

Project Proposal

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TradeM8 - Your Trading Assistant 1

Abstract

Financial markets are a big part of the world today, with trillions of dollars changing hands every day. Crypto and stock markets stand out as especially unpredictable ones. Traders depend on detailed methods like Technical Analysis (TA) and Fundamental Analysis (FA) to navigate them. But with various tools spread out online, each offering different features, it's tough for traders to handle the analysis smoothly. Spotting chart patterns in TA takes a lot of practice, and FA means keeping up with constant news updates. Thanks to progress in Artificial Intelligence (AI), we can now use advanced models to handle these tasks automatically. Our project, TradeM8, will automate TA by detecting patterns and applying technical indicators. It will also cover FA through sentiment analysis on financial news and articles. This all-in-one setup will make things easier for traders, cutting down on manual work.

1. Introduction

Trading has been part of human life since the bartering systems of ancient civilizations like Egypt. The modern stock industry appeared in 1602 with the establishment of the Amsterdam Stock Exchange. Evolving technology later gave rise to electronic trading in the late 20th century, which increased global participation. Today, electronic trading markets shape the world with trillions of dollars traded daily.

Nowadays, traders can buy and sell electronic shares online, but this comes with the steep learning curve of TA and FA, as these markets are highly volatile. For TA, a trader has to analyze chart patterns and read price action, while for FA, they must stay up to date with news and articles. Often, tools for TA and FA are scattered, and traders have to look for them separately. For example, for news, they must constantly check financial websites, which can be time-consuming and tedious.

With the help of AI, we are now approaching trading techniques where a trader does not need to learn the entire steep curve; instead, basic knowledge combined with automated TA and FA tools can provide the necessary insights.

This project, TradeM8, helps traders with TA and FA by drawing chart patterns and performing sentiment analysis on news. As a result, they no longer have to manually draw patterns (including head-and-shoulders and double-top) or constantly monitor news feeds.

2. Goals and Objectives

Our project's objectives include:

- Detect well-known chart patterns (e.g. Head & Shoulders, Rising Channel) using deep learning.
- Validate detected signals using technical indicators (EMA, RSI, ADX).
- Provide real-time market data via Trading APIs.

- Perform sentiment and fundamental analysis using financial news sources (e.g. Yahoo Finance, CNN Crypto).
- Reduce false trading signals through indicator-based and sentiment-informed filtering.

3. Scope of the Project

TradeM8 will help the user in decision making using automated TA and FA using Artificial Intelligence. This application fetches live data from Exchange APIs and constructs

TradeM8 - Your Trading Assistant 2

candlestick charts. Then it calculates metrics such as Exponential Moving Averages (EMA), Relative Strength Index (RSI) etc that are helpful in overall decision making. For TA, we will be using deep learning models to detect chart patterns and validate them using indicators such as Exponential Moving Averages (EMA), Relative Strength Index (RSI) etc. FA will be automated by applying sentiment analysis on updates and news.

4. Initial Study and Work Done so Far

As far as our research is concerned, a number of attempts have been made to automate Technical Analysis in digital asset markets using deep learning and AI. We found that newer visual methods, like those based on YOLOv8, can generate candlestick patterns and trading signals with 84-87% accuracy. Other work shows convolutional neural networks (CNNs) are good at pulling temporal features from candlestick charts, leading to better short-term forecasts. This highlights how image-based deep learning can work well for TA. Meanwhile, time-series models like LSTM have been used on market data to track price sequences and boost prediction reliability. Even basic candlestick patterns, such as Doji and Hammer, offer value when paired with neural networks. Research also points out that just detecting patterns is not sufficient—adding technical indicators like EMA and RSI helps weed out anomalies and steady the predictions. Combining market indicators with on-chain details, like transaction activity, can lead to strong trend predictions too. That said, plenty of deep learning models perform adequately on historical data but struggle with future moves due to overfitting and market noise. These insights led us to develop TradeM8 as a hybrid tool that brings together TA and FA. It combines pattern detection, indicator checks, and sentiment analysis from financial news into a real-time helper for traders

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TradeM8 - Your Trading Assistant 3